

Exp : 5 VPS / Virtual Machine Setup and Linux Configuration

Aim

To provision a Virtual Private Server (VPS) or create a virtual machine using virtualization software such as **VirtualBox** or **VMware Workstation**, install a Linux operating system, and configure network settings including IP address, subnet mask, gateway, and DNS.

Theory

A **Virtual Private Server (VPS)** is a virtualized environment that acts like a dedicated server. It can be hosted on cloud platforms or created locally using virtualization tools.

Virtualization software such as:

- VirtualBox
- VMware Workstation

allows running multiple operating systems on a single physical machine.

Linux distributions like:

- Ubuntu
- CentOS
- Debian

are commonly used due to stability, security, and open-source nature.

Procedure

Step 1: Install Virtualization Software

Option A: Install VirtualBox

1. Download from official website
2. Install with default settings

Option B: Install VMware Workstation

1. Download installer
2. Follow installation wizard

Step 2: Create Virtual Machine

In VirtualBox:

1. Open VirtualBox → Click **New**
2. Enter:
 - Name: LinuxVM
 - Type: Linux
 - Version: Ubuntu (64-bit)
3. Allocate:
 - RAM: 2 GB (minimum)
 - CPU: 1–2 cores
4. Create Virtual Hard Disk:
 - Type: VDI
 - Size: 20 GB

Step 3: Attach ISO File

1. Download ISO (e.g., Ubuntu)
2. Go to VM Settings → Storage
3. Attach ISO file

Step 4: Install Linux OS

1. Start VM
2. Select **Install Ubuntu**
3. Follow steps:
 - Language selection
 - Keyboard layout
 - Installation type (Normal)
 - Create username & password
4. Click **Install**
5. Restart VM after installation

Step 5: Configure Network (Automatic - DHCP)

Check IP:

```
ip a
```

Check connectivity:

```
ping google.com
```

Step 6: Configure Static IP Address

Edit Netplan config:

```
sudo nano /etc/netplan/01-netcfg.yaml
```

Example configuration:

```
network:
```

```
  version: 2
```

```
  ethernets:
```

```
    enp0s3:
```

```
      dhcp4: no
```

```
      addresses: [192.168.1.100/24]
```

```
      gateway4: 192.168.1.1
```

```
      nameservers:
```

```
        addresses: [8.8.8.8, 8.8.4.4]
```

Apply changes:

```
sudo netplan apply
```

Step 7: Verify Network Configuration

Check IP:

```
ip a
```

Check gateway:

```
ip route
```

Test DNS:

```
ping google.com
```

Step 8: (Optional) VPS on Cloud

Instead of local VM, you can use:

- Microsoft Azure
- Amazon EC2

Steps:

1. Create VM instance
2. Select Linux image
3. Configure networking
4. Connect via SSH

Result

Successfully created a virtual machine, installed Linux OS, and configured network settings including IP address, gateway, and DNS.